

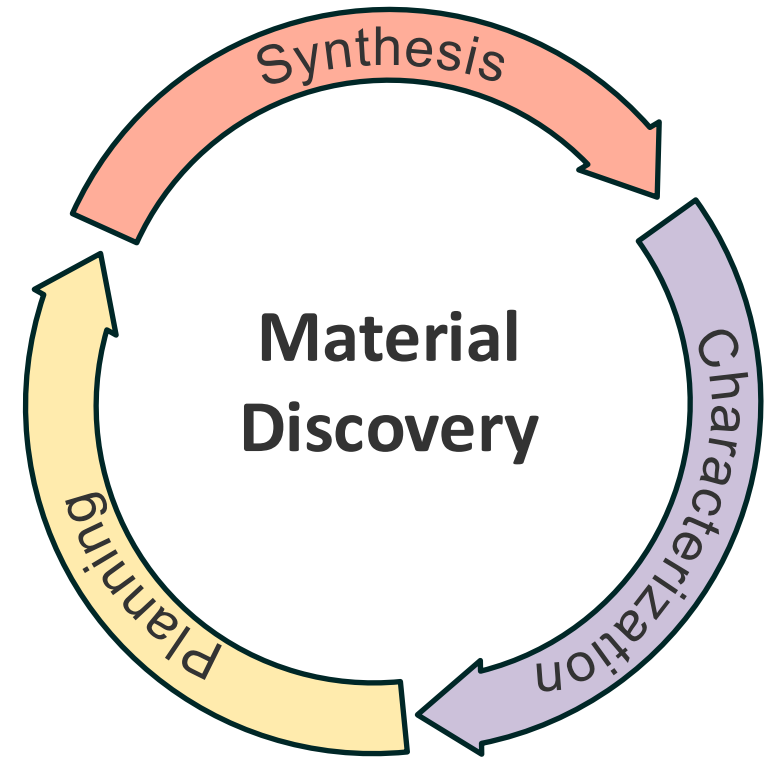
AE Microscopy for Material Exploration: scalable material discovery

Boris Slautin

University of Tennessee, Knoxville



- The bottleneck is shifting from making materials to characterizing, interpreting, and acting on them.
- To close the materials discovery loop, characterization must become automated, adaptive, and connected to real-time decision-making.



1. Close the loop:

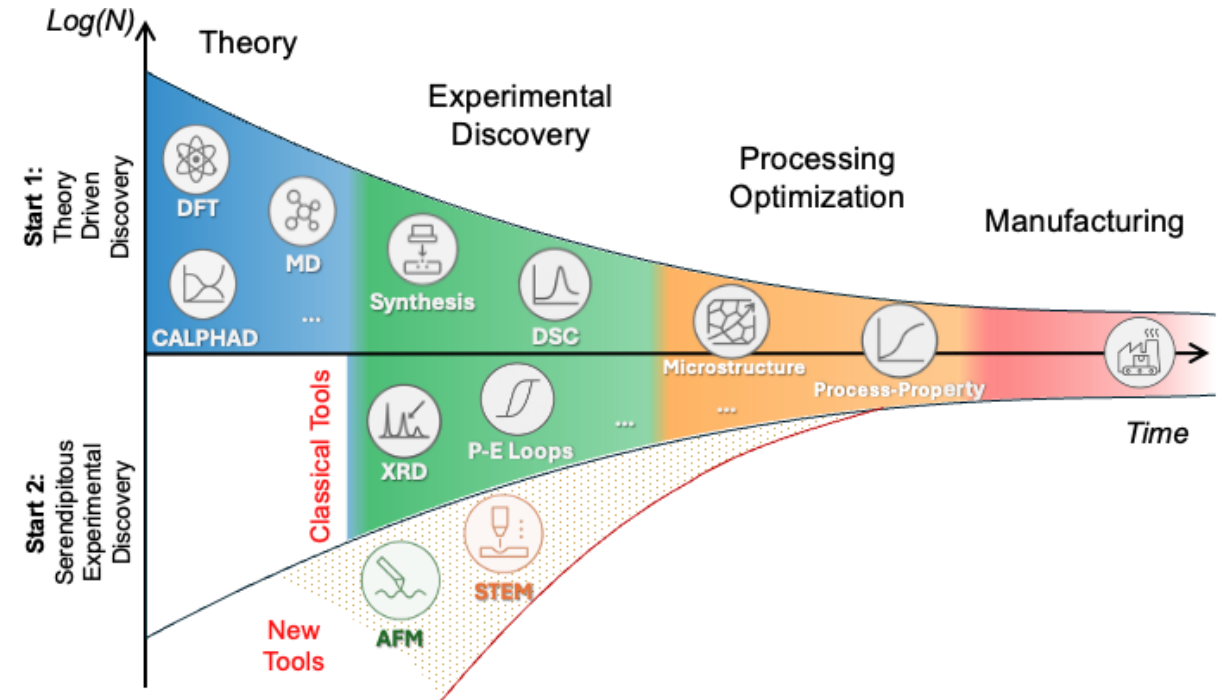
To accelerate materials discovery, we have to close the loop from synthesis to characterization and model update

2. Explore smarter:

ML/AI methods along with the predictive theory allow better initial choices; however, most materials are discovered serendipitously even now

3. New tool paradigms:

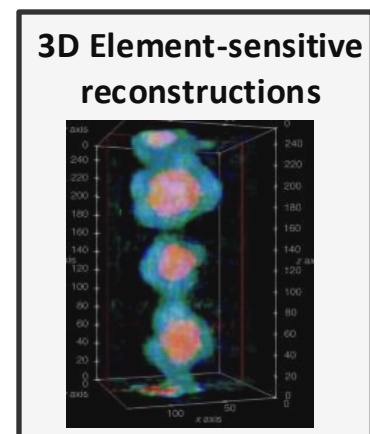
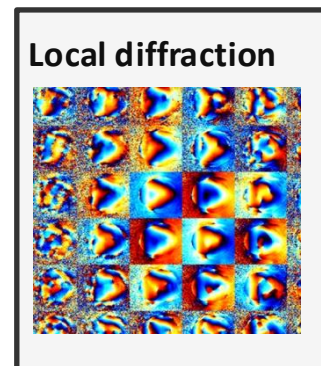
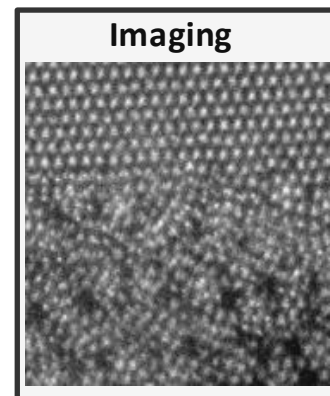
Characterization tools are often limited by fundamental physics: we need different tool paradigms!



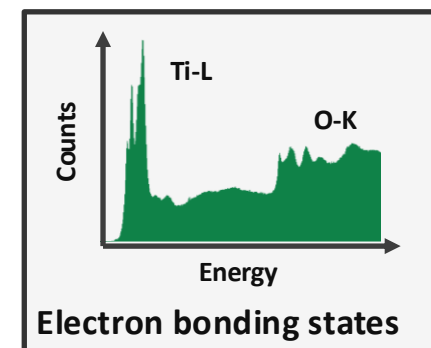
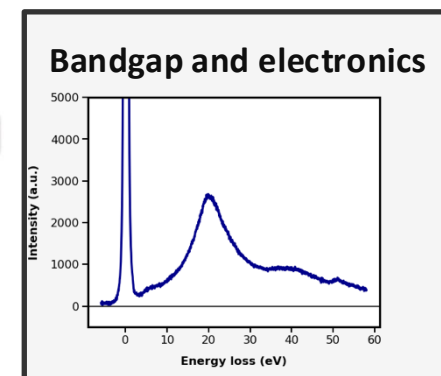
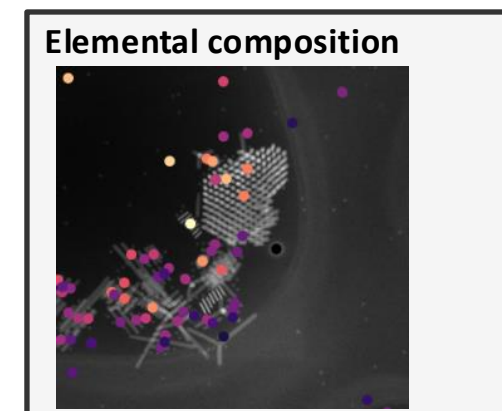
Multimodality

Macro	Micro
X-Ray	Electron crystallography
Raman	EELS
CHemistry	EDS
ARPES	Core loss EELS

- **Strong** and **weak** proxies for synthesis and combinatorial libraries
- **Correlative structure-property relationships**
- Structure for **non-crystalline** materials
- Microstructure information is crucial for **structure-processing** relationships
- In many cases, we can expect to learn **generative physical model**



ThermoFisher Spectra 300

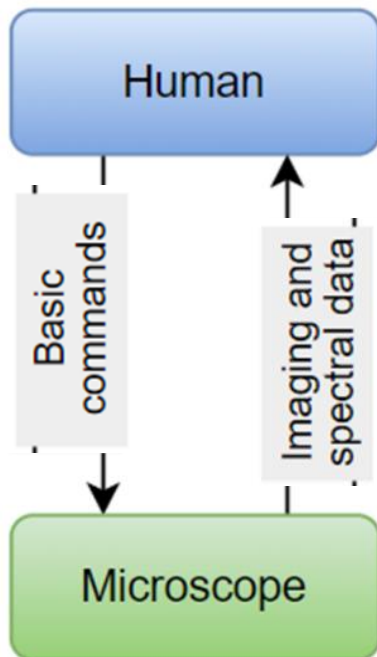


R. Hovden

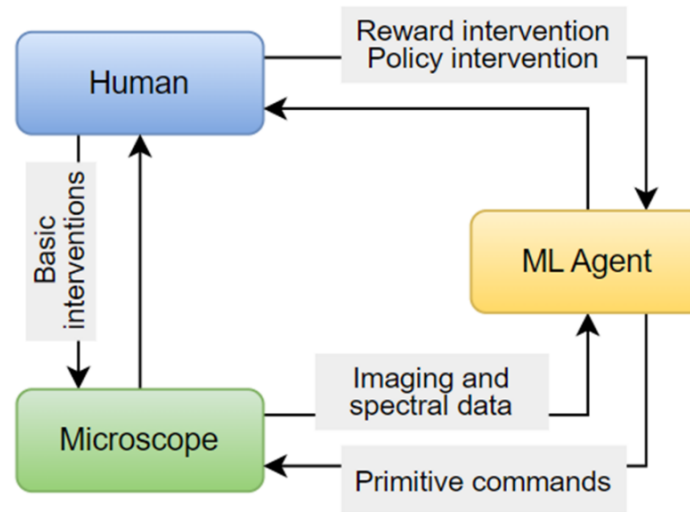
A. Lupini

Bottleneck 1: Low-level operational human control

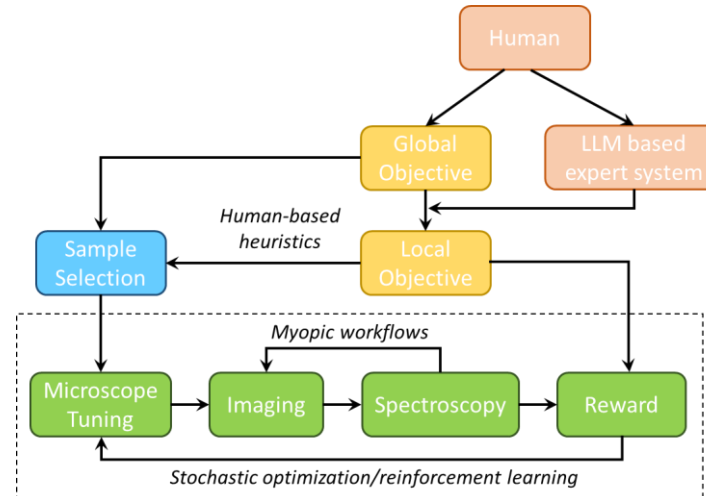
Current generation of tools



hAE Workflows

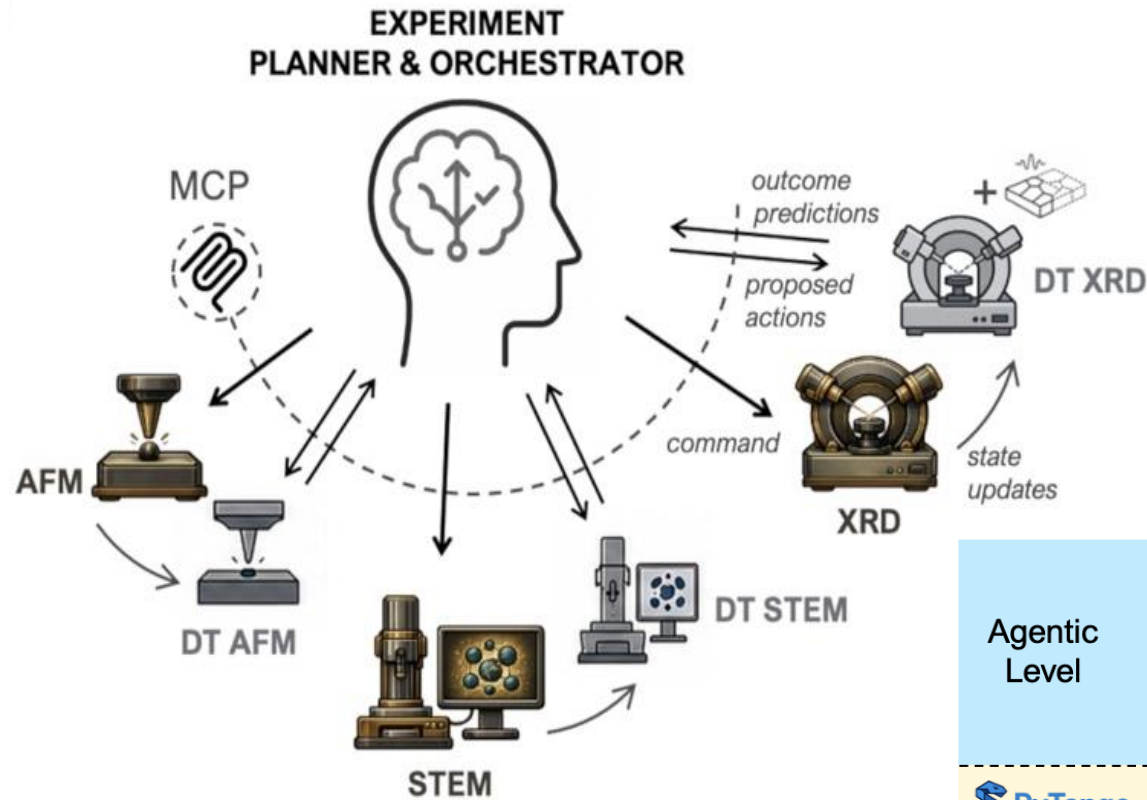


LLM based co-scientists

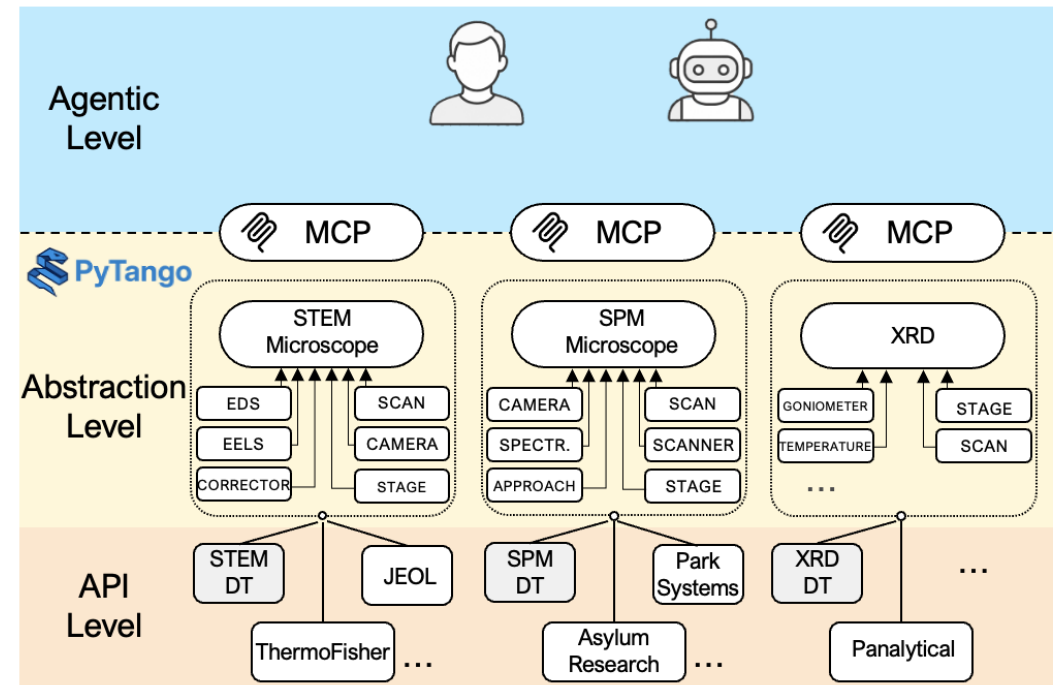


Reducing human involvement in low-level instrument operation

Bottleneck 2: Fragmented multimodal orchestration



Agentic multimodal
integration

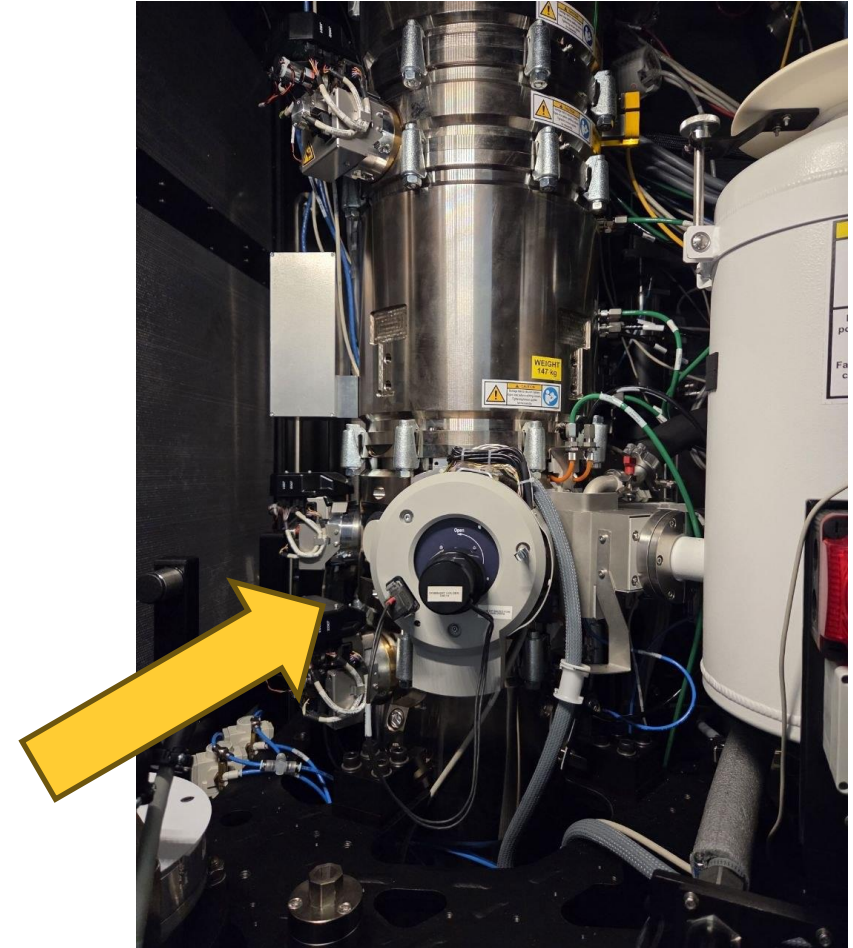



pycroscopy
Asyncroscopy

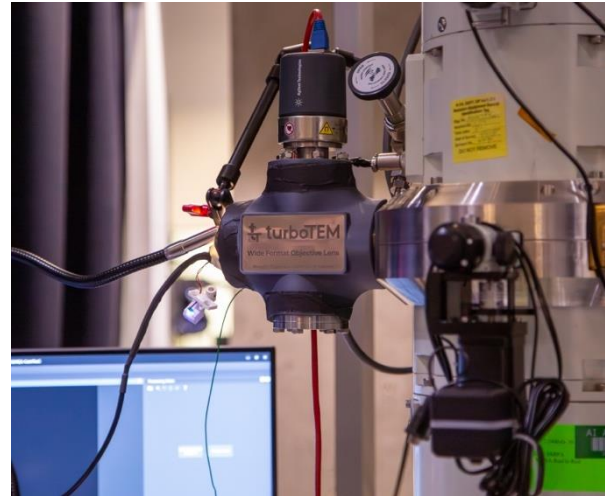


Bottleneck 3: One sample at a time

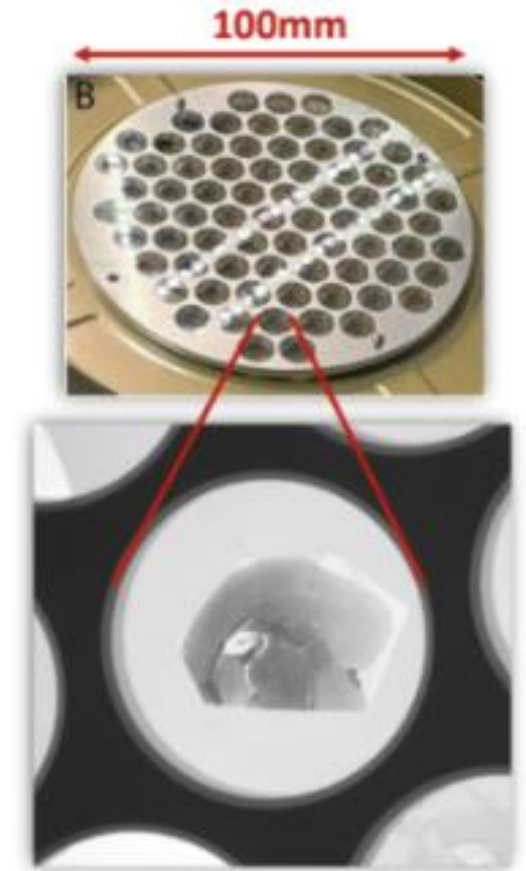
One specimen in the column at a time, with exchange and realignment between each.



- **Horizon** wide-format objective lens
- Multiorfices wafer (100 mm)



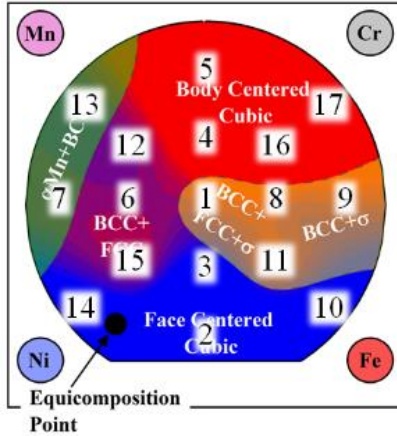
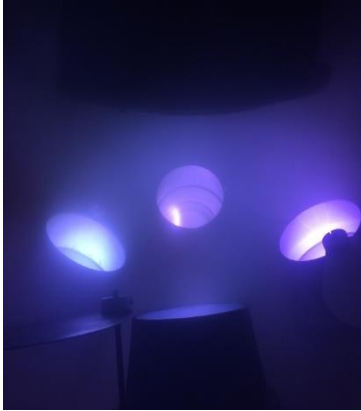
Horizon wide-format objective lens installed at customer site – Allen Institute for Brain Science. Version shown is for the JEOL 1200 platform. Photo credit: Jenny Burns, Allen Institute



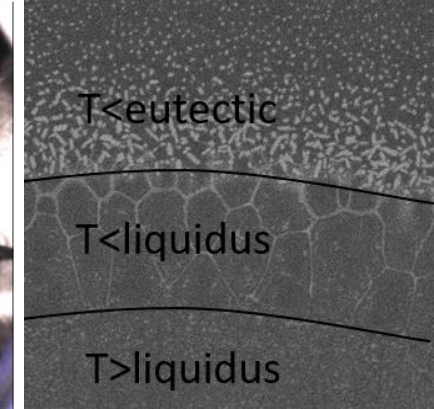
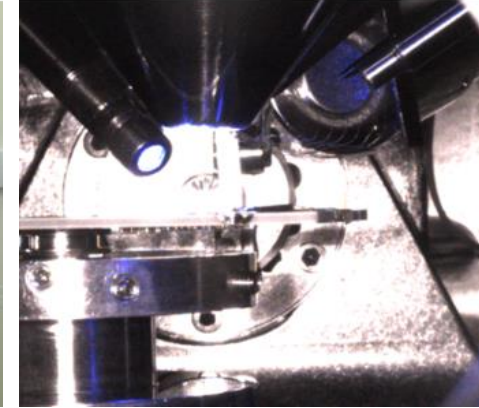
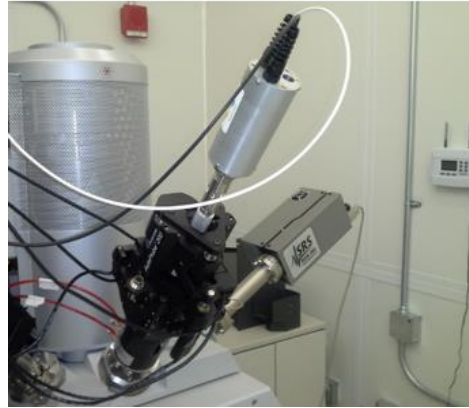
100mm multi orifice wafer. Image credit: Kim Gruver, Allen Institute for Brain Science.

Automated Experiment At UT Knoxville

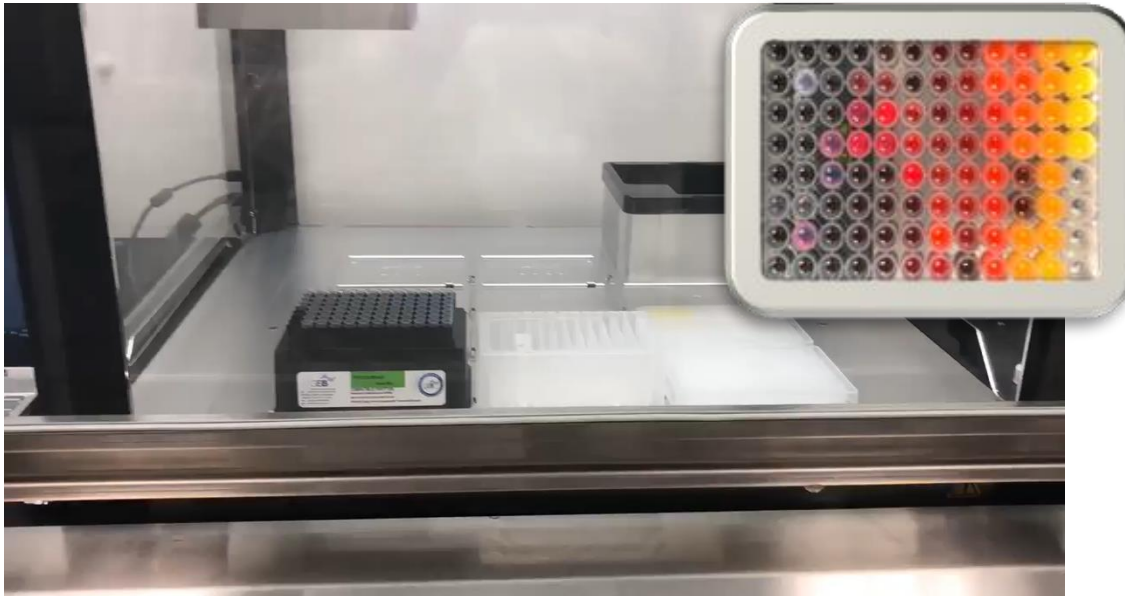
Combinatorial growth



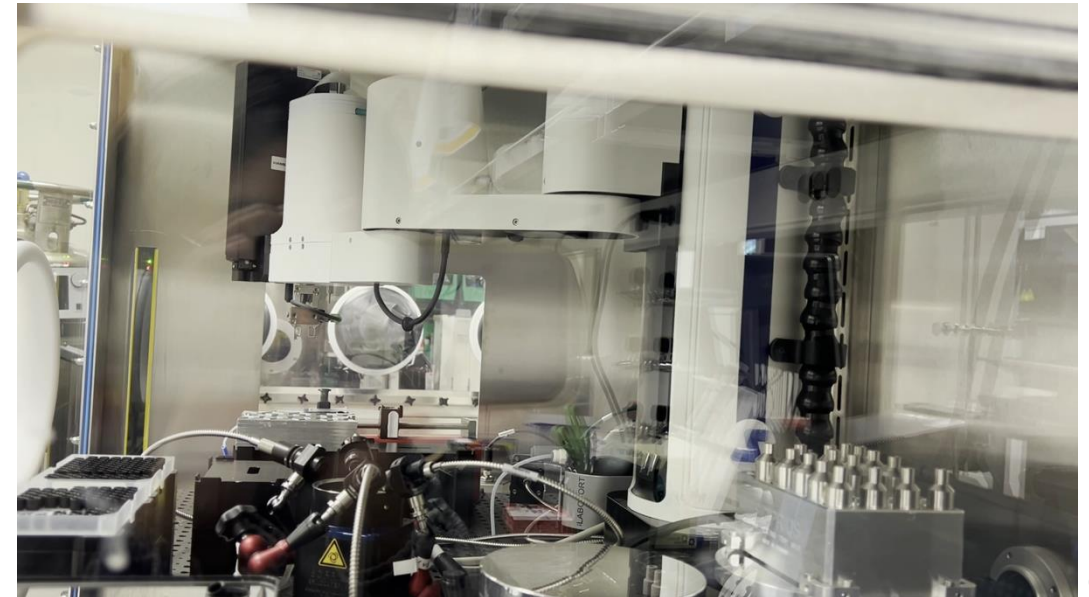
Combinatorial laser processing



High-throughput synthesis



Robotic film deposition/characterization



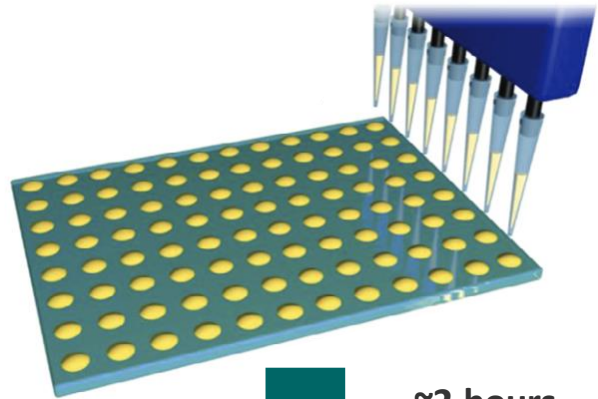
Phil Rack



Mahshid Ahmadi

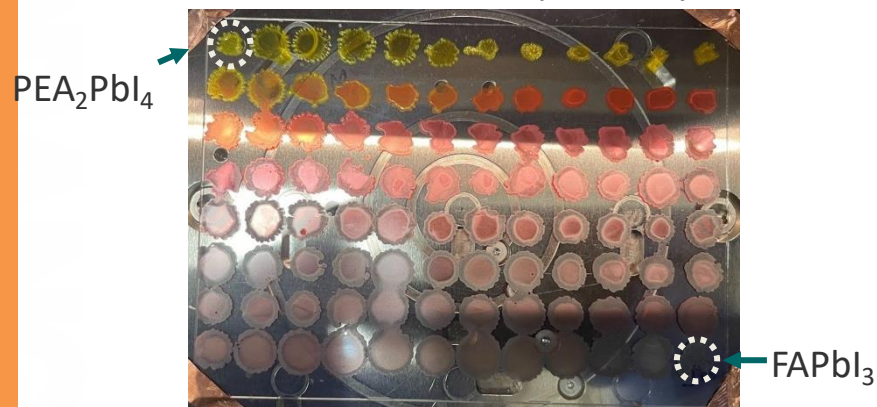
Photoactive Materials Characterization

High-throughput precursor drop-casting synthesis

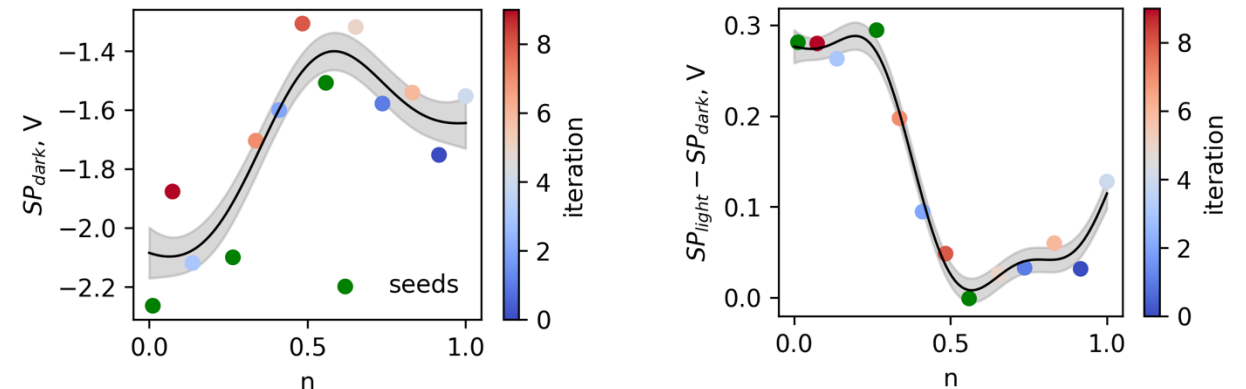
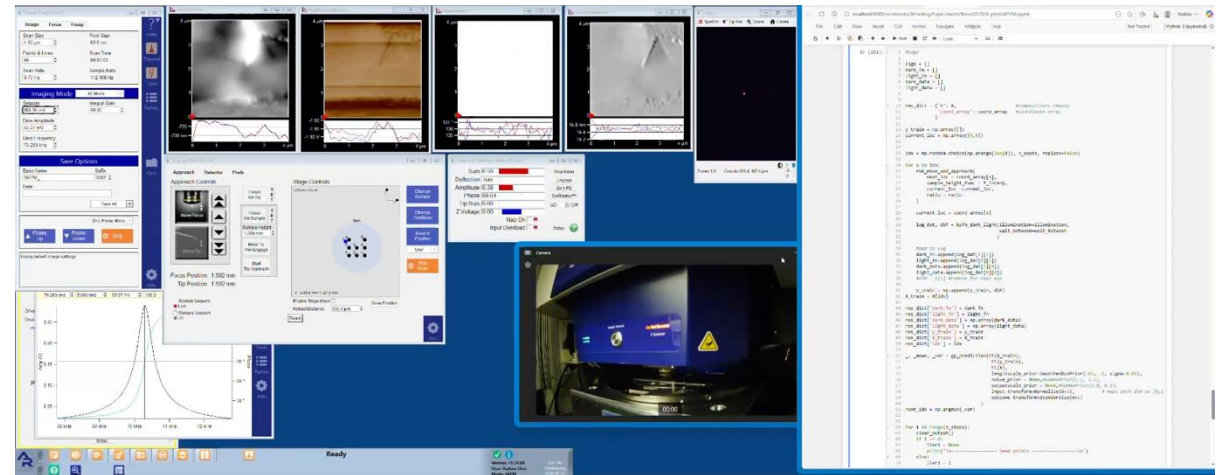


~2 hours

Quasi-2D MHP droplet array



96-sample combinatorial library



- The entire library was characterized within ~1.5 h
- Photo-CPD is used to estimate photoactivity

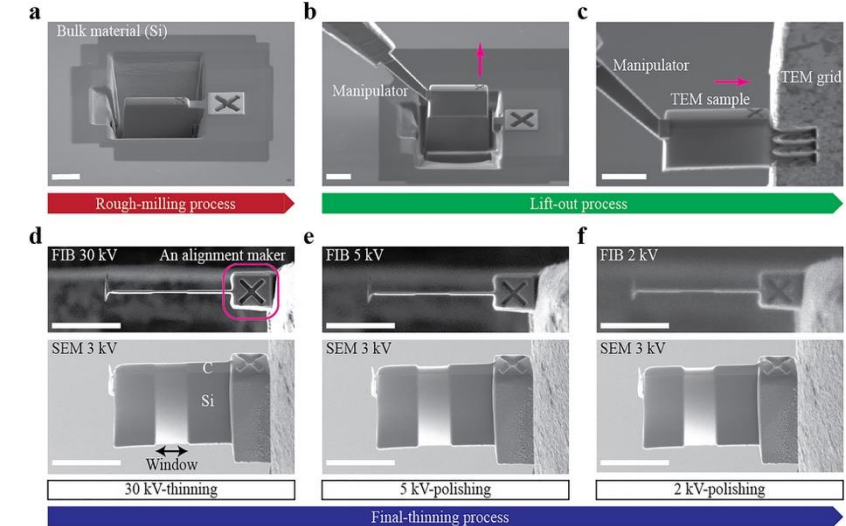
Robotic fabrication of lamellae

Article | [Open access](#) | Published: 03 November 2021

Robotic fabrication of high-quality lamellae for aberration-corrected transmission electron microscopy

[Hideyo Tsurusawa](#) ✉, [Nobuto Nakanishi](#), [Kayoko Kawano](#), [Yiqiang Chen](#), [Mikhail Dutka](#), [Brandon Van Leer](#) & [Teruyasu Mizoguchi](#) ✉

[Scientific Reports](#) 11, Article number: 21599 (2021) | [Cite this article](#)

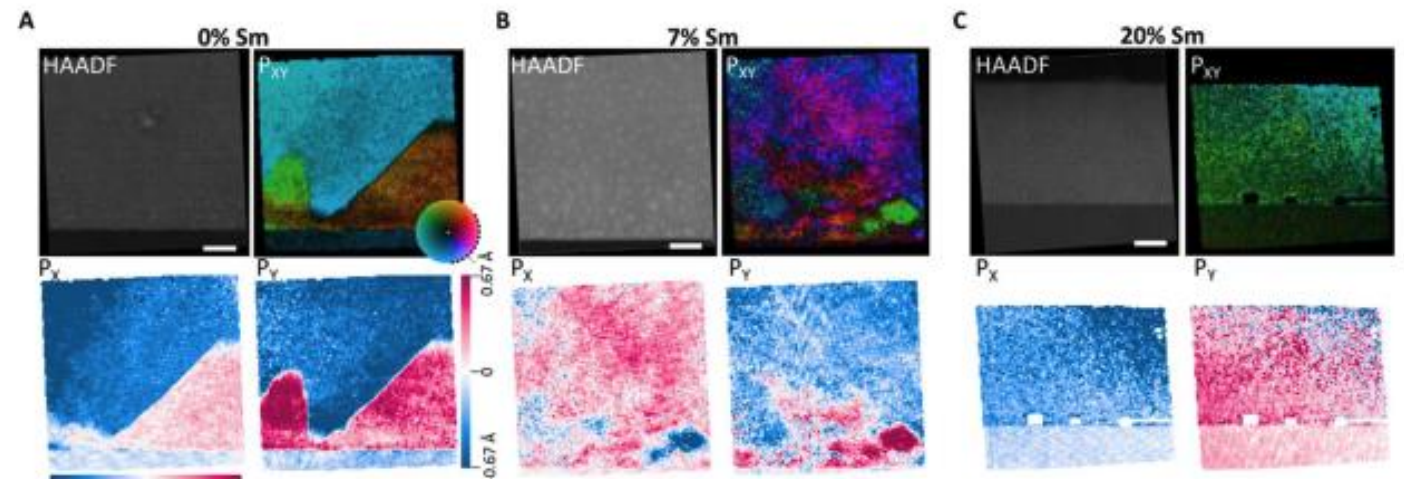


Article | [Open access](#) | Published: 21 August 2020

Causal analysis of competing atomistic mechanisms in ferroelectric materials from high-resolution scanning transmission electron microscopy data

[Maxim Ziatdinov](#), [Christopher T. Nelson](#), [Xiaohang Zhang](#), [Rama K. Vasudevan](#), [Eugene Eliseev](#), [Anna N. Morozovska](#), [Ichiro Takeuchi](#) & [Sergei V. Kalinin](#) ✉

[npj Computational Materials](#) 6, Article number: 127 (2020) | [Cite this article](#)

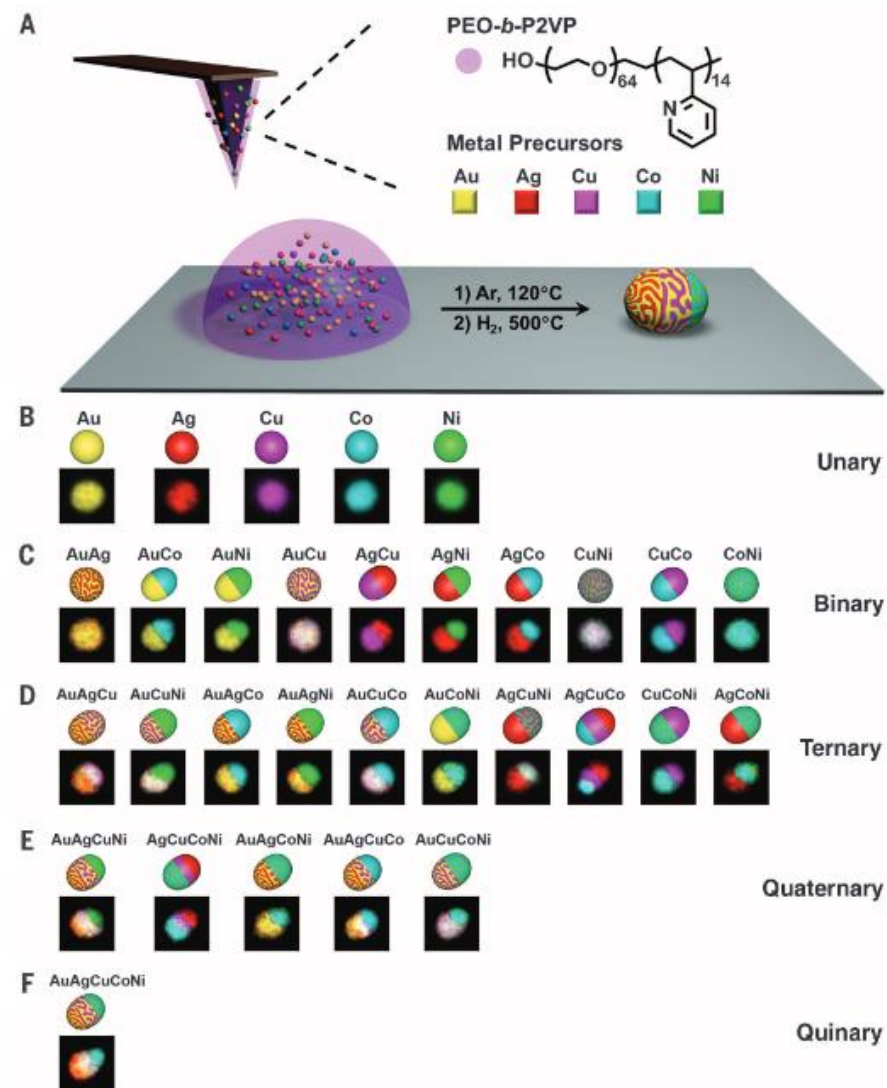


Sm-BFO combinatorial library

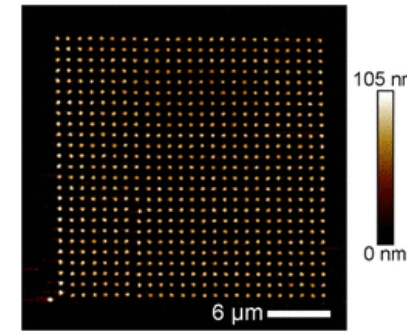
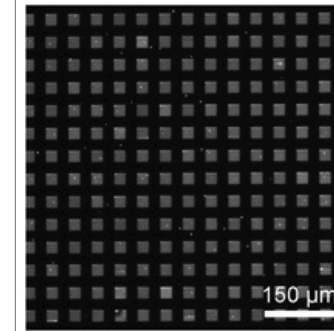
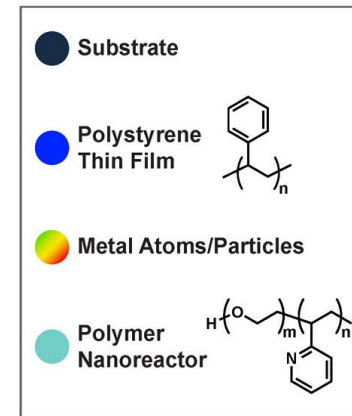
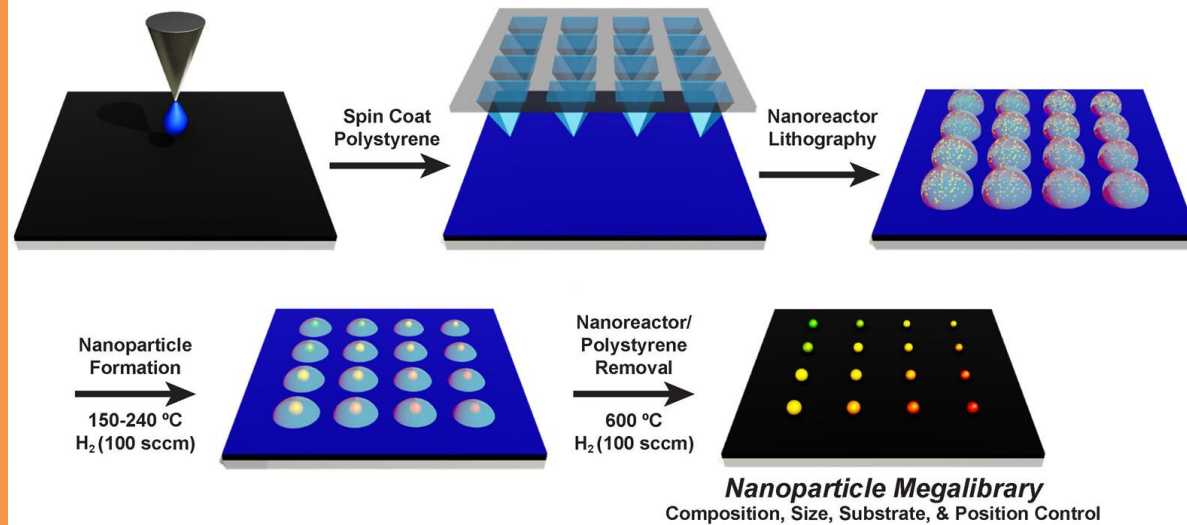
Polyelemental nanoparticle libraries

Peng-Cheng Chen,^{1,2} Xiaolong Liu,³ James L. Hedrick,^{2,4} Zhuang Xie,^{2,5}
 Shunzhi Wang,^{2,5} Qing-Yuan Lin,^{1,2} Mark C. Hersam,^{1,3,5}
 Vinayak P. Dravid,^{1,2} Chad A. Mirkin^{1,2,4,5*}

- **Scanning-probe block copolymer lithography** prints metal-precursor-loaded polymer nanoreactors; after annealing, each nanoreactor converts into a multimetallic nanoparticle.
- The ordered nanoparticle library can be directly mapped STEM making it a strong platform for automated screening of combinatorial nanomaterials.

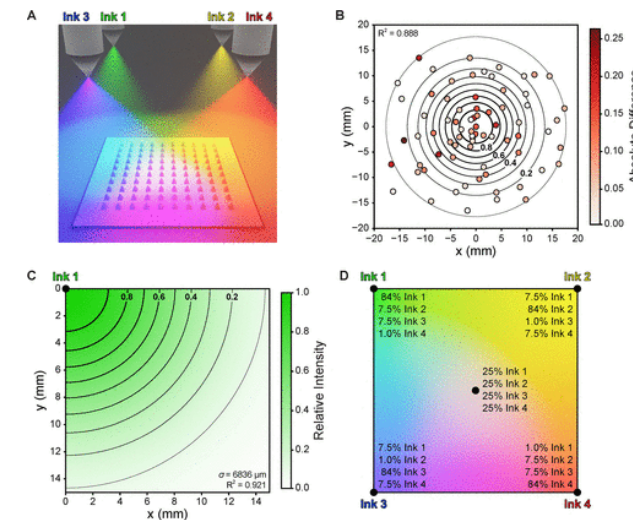


Combinatorial libraries -> partical megalibraries



Representative material systems:

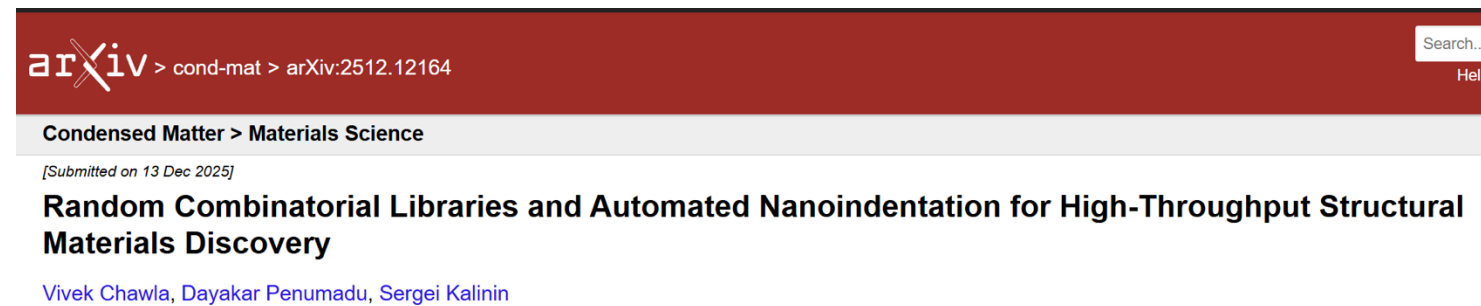
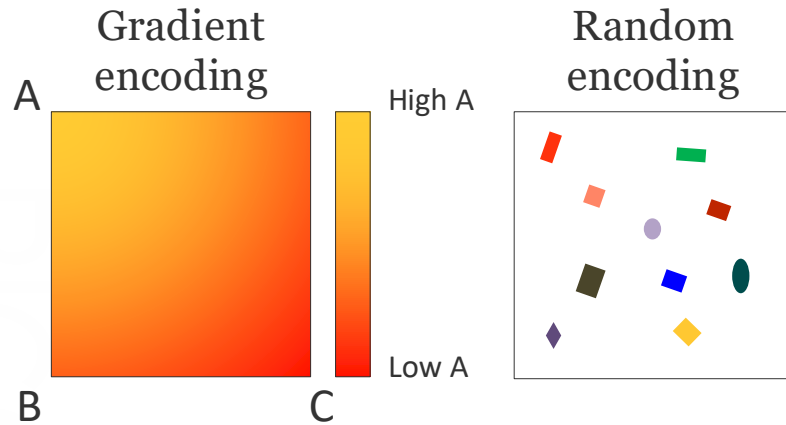
- metal/alloy nanoparticles
- polyelemental particles
- catalyst libraries
- Heterostructures
- mixed-halide perovskites



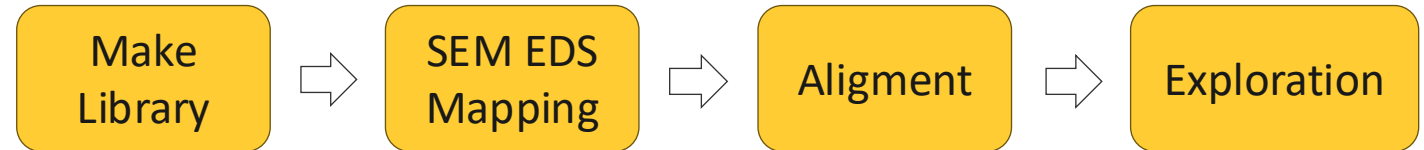
>10-million-particle libraries through polymer pen lithography (PPL)

Random Libraries

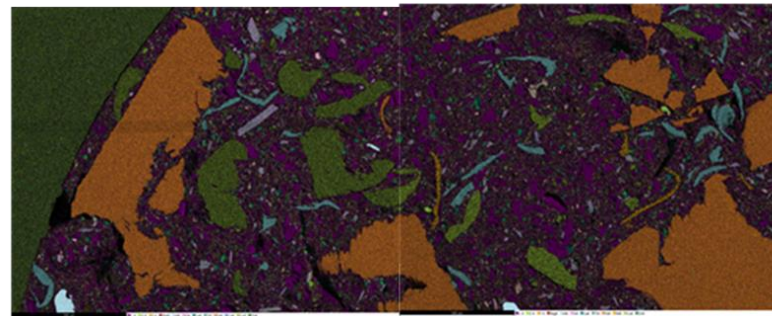
“Random library” a single specimen that contains many discrete particles/inclusions, where each particle is treated as a separate material/composition point.



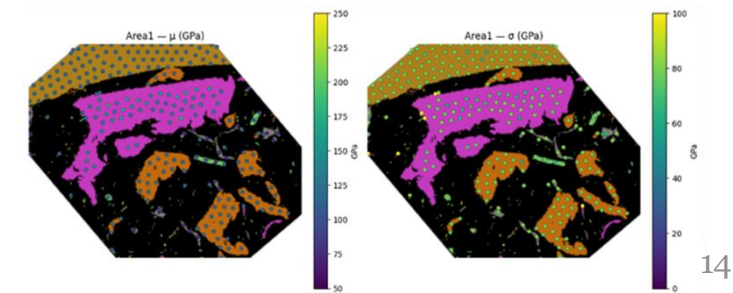
Random Library Construction



Chemical Space Exploration

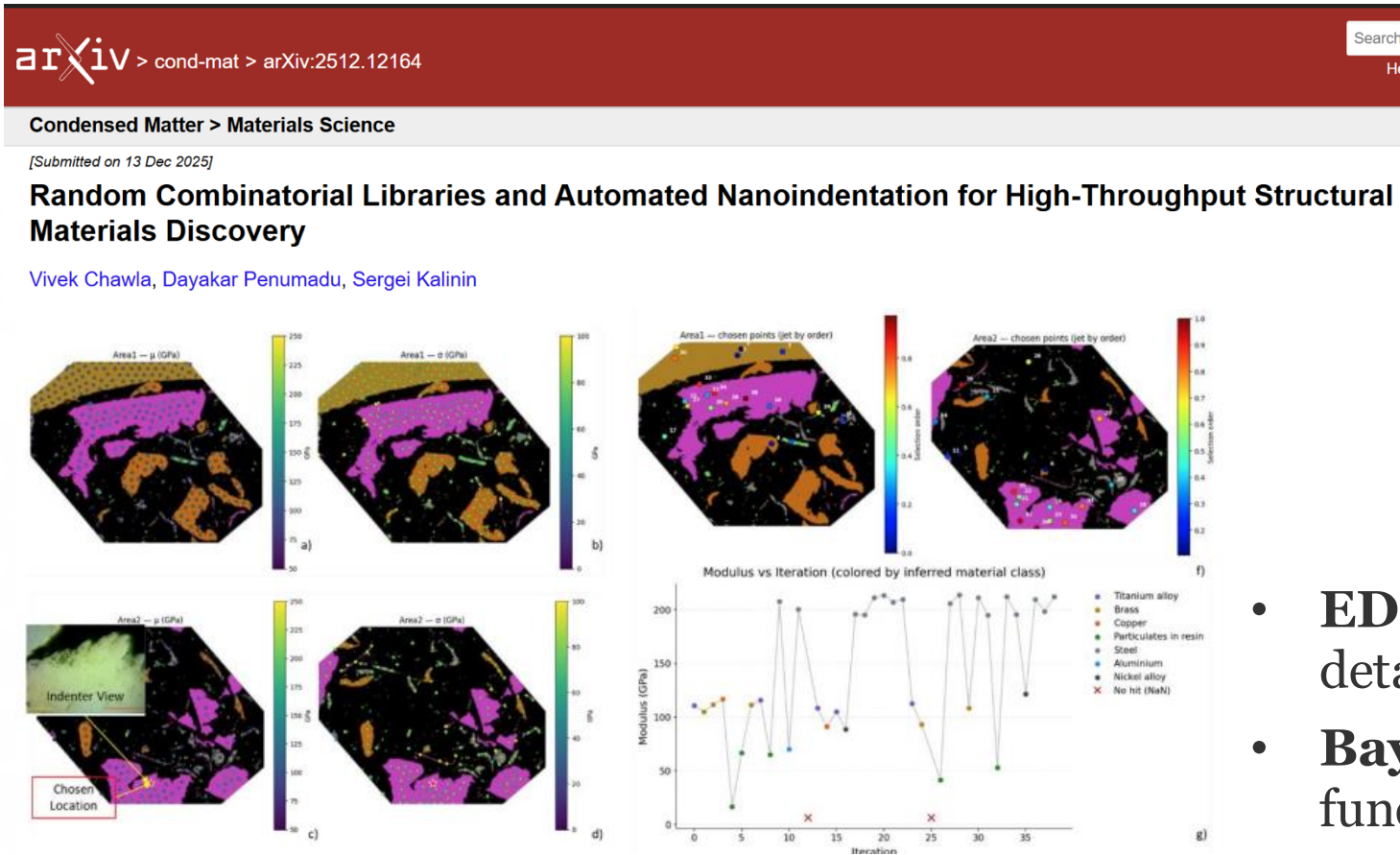


Automated Nanoindentation

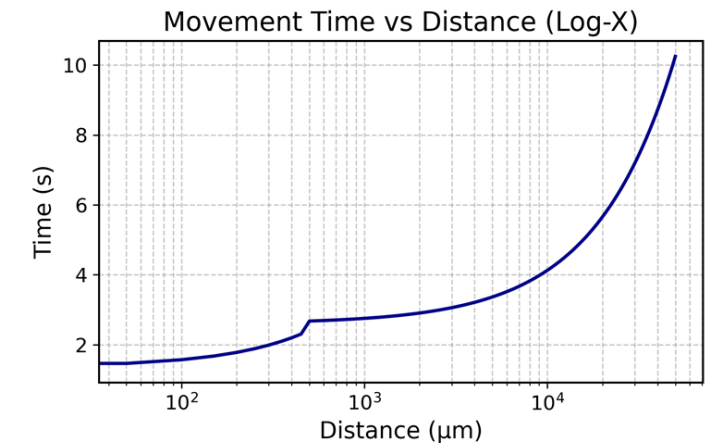


Random Libraries

“Random library” a single specimen that contains many discrete particles/inclusions, where each particle is treated as a separate material/composition point.



Cost function and targeting



- **EDS mapping:** fast + clustering or detailed local composition.
- **Bayesian Optimization** with cost function integration

- Many materials in one run – a single specimen yields modulus for Ti, steel, brass, Cu, Al and Ni alloys – without human intervention.

Random Libraries for Mechanical Properties

arXiv > cond-mat > arXiv:2512.12164

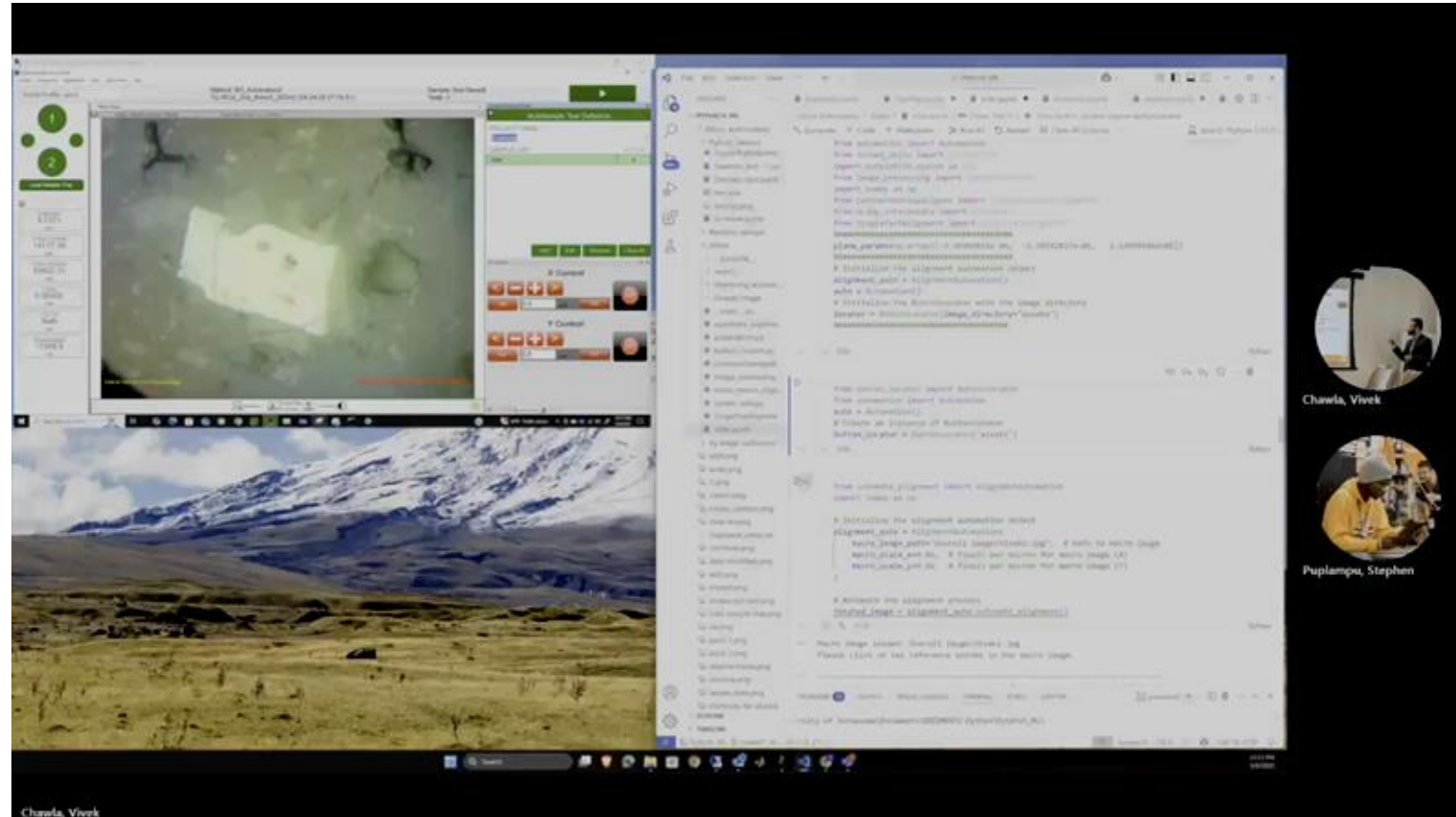
Search...
Help

Condensed Matter > Materials Science

[Submitted on 13 Dec 2025]

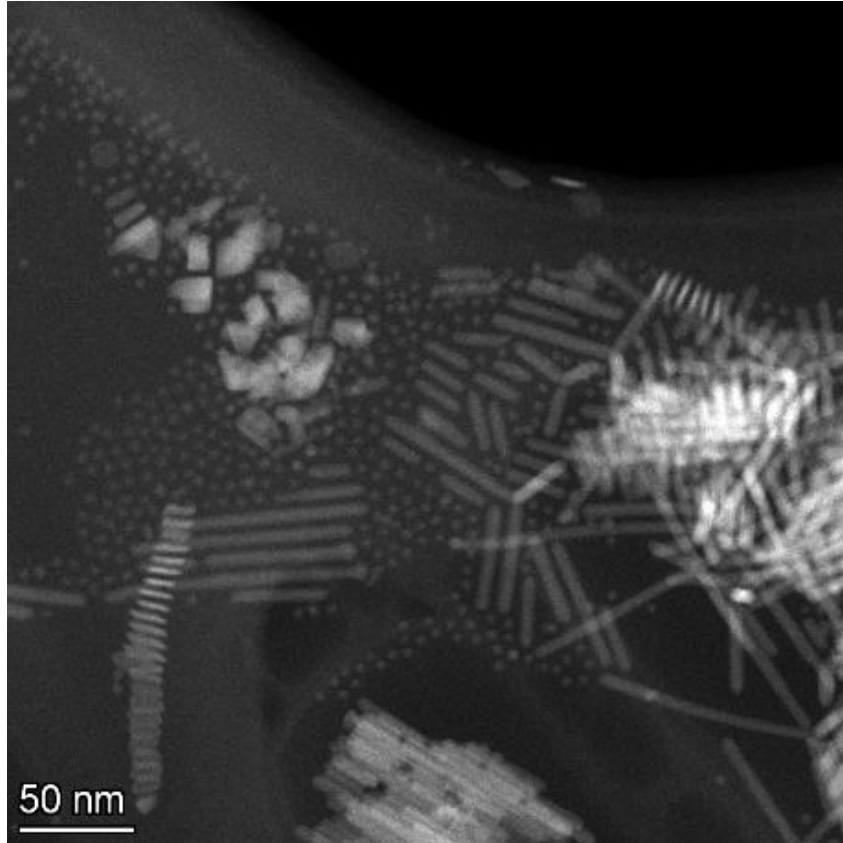
Random Combinatorial Libraries and Automated Nanoindentation for High-Throughput Structural Materials Discovery

Vivek Chawla, Dayakar Penumadu, Sergei Kalinin



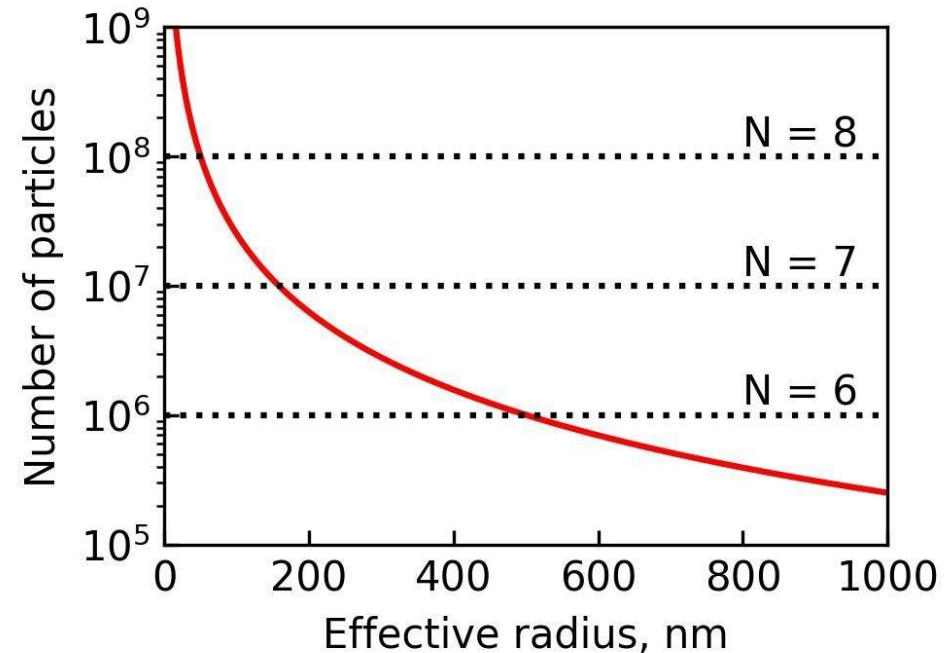
Acceleration factor > 100

Random Libraries for STEM



CdSe, CdS quantum dots, CdSe nanosheetlets, CdS nanorods, InP-based nanoparticles

Naïve lower-bound estimate: random particle libraries span **at least ~6-8** effective **dimensions** per standard TEM grid



- Random libraries enable dense multiplexing of composition, phase, and microstructure within a single specimen.
- STEM-based chemical fingerprinting allows local identification and characterization without prior positional encoding.

Random Libraries for STEM

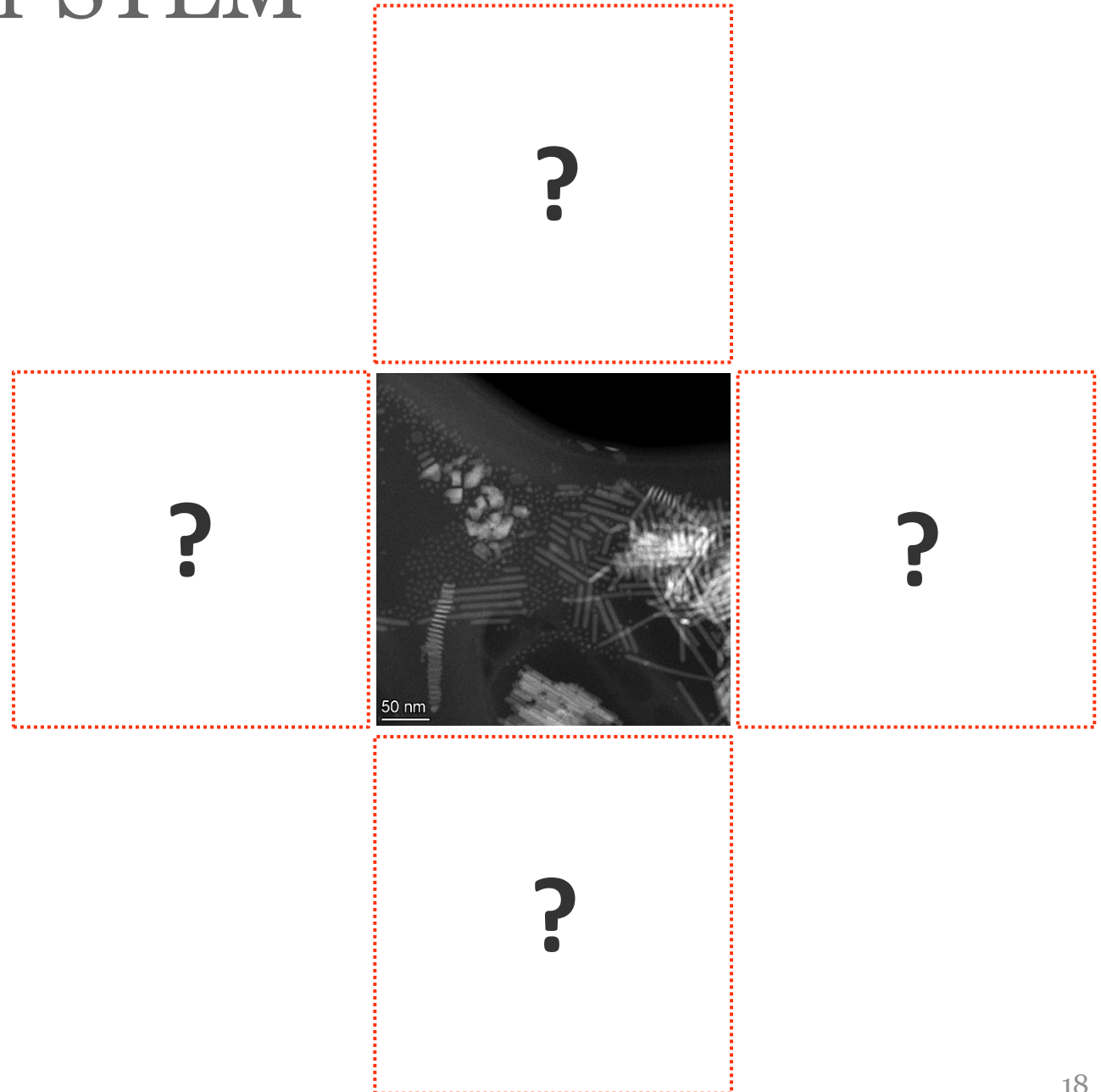
Limited field of view: everything outside the observed region is hidden by a “fog of war.”



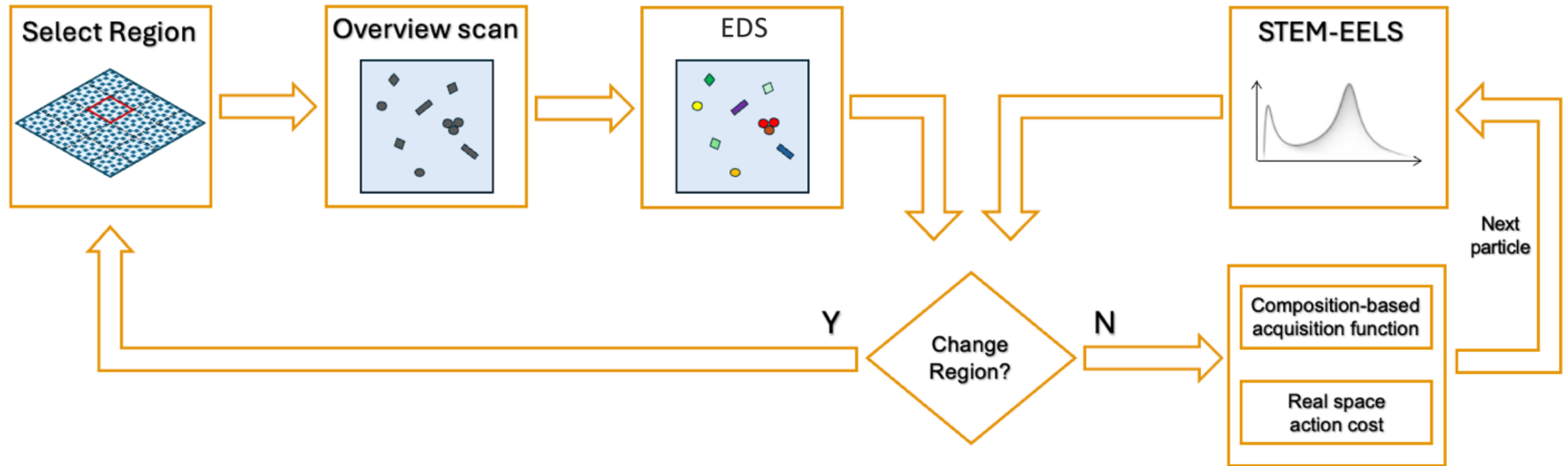
Sparse compositional access from limited STEM field of view.



Heroes of Might and Magic III

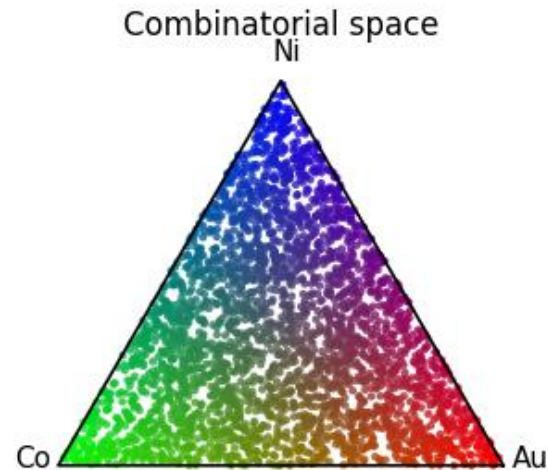
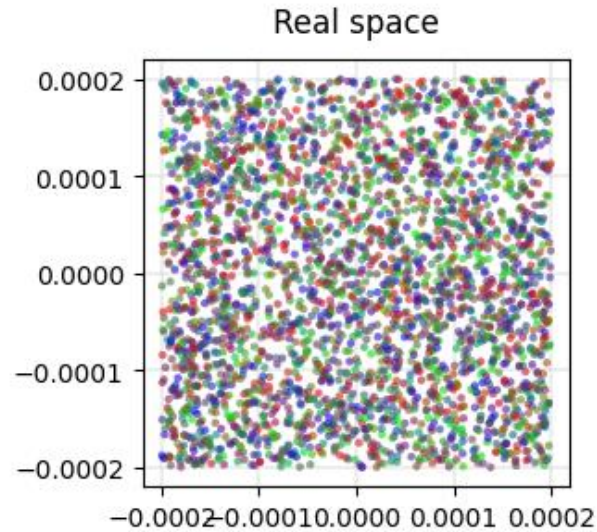


STEM for Random Libraries



- Overview imaging identifies particle locations within the selected region
- EDS provides local compositional fingerprints for all particles within the FoV, defining the accessible compositions.
- A BO-driven loop uses EELS functional measurements on particles selected via cost-aware, composition-based acquisition, balancing information gain against motion cost.
- After each iteration, a region-switch decision is made to balance the remaining local opportunity against the cost of stage motion.

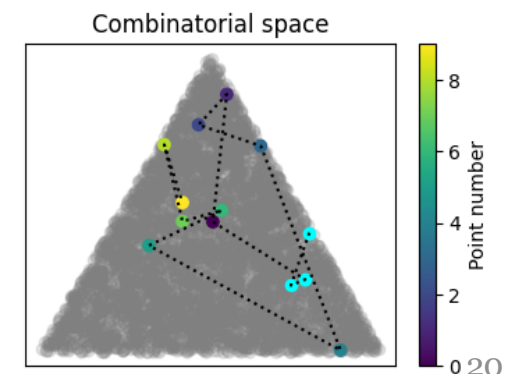
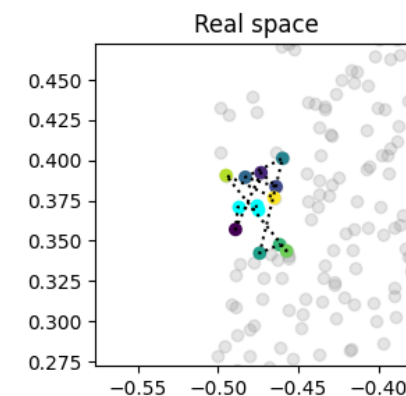
Simulation: STEM for Random Libraries



Constant **motion cost** within beam shift; linear increase beyond.

$$c_{\text{move}}(d) = \begin{cases} c_{\text{intro}}, & d \leq d_{\text{th}}, \\ \alpha(t_0 + \frac{d}{v}), & d > d_{\text{th}}, \end{cases}$$

- **Cost-aware acquisition:** BO is defined in compositional space, but weighted by real-space motion cost, so the algorithm favors chemically informative particles that are also experimentally cheap to reach.
- At each iteration, the algorithm balances measuring known particles within the current FOV against moving to unexplored regions to improve compositional coverage.



STEM for Random Libraries: Decision making



- If many good unmeasured particles remain → **stay**
- If only weak or repetitive particles remain → **move**

STEM for Random Libraries: Decision making

- **Expected gain** from exploring of **new region**



- The **remaining opportunity** within the current **field of view**

Remaining particle part:

$$R = 1 - \frac{N_{\text{meas}}}{N_{\text{tot}}}$$

Maximum acquisition function:

$$G(i) = \max_{i \in \mathcal{U}_{\text{unm}}} a_t(c_i)$$

$$V_{\text{stay}}(i) = R \cdot G(i)$$

STEM for Random Libraries: Decision making

- **Expected gain** from exploring of **new region**

Number of particles:

$$N \sim \text{Poisson}(\rho A)$$

Potential switch gain:

$$G_{\text{new}}(i) = \max_{j=1,\dots,N} a_i(c_j)$$

$$E_{\text{gain}}(i) = \mathbb{E}[G_{\text{new}}(i)]$$

Cost of switching:

$$C_{\text{switch}}(t) = t_{\text{switch}} \cdot \frac{V_{\text{stay}}(t)}{t_{\text{meas}}}$$



acquisition
function

- The **remaining opportunity** within the current **field of view**

Remaining particle part:

$$R = 1 - \frac{N_{\text{meas}}}{N_{\text{tot}}}$$

Maximum acquisition function:

$$G(i) = \max_{i \in \mathcal{U}_{\text{unm}}} a_t(c_i)$$

$$V_{\text{stay}}(i) = R \cdot G(i)$$

STEM for Random Libraries: Decision making

- **Expected gain** from exploring of **new region**



- The **remaining opportunity** within the current **field of view**

Cost of switching:

$$C_{\text{switch}}(t) = t_{\text{switch}} \cdot \frac{V_{\text{stay}}(t)}{t_{\text{meas}}}$$

Expected gain:

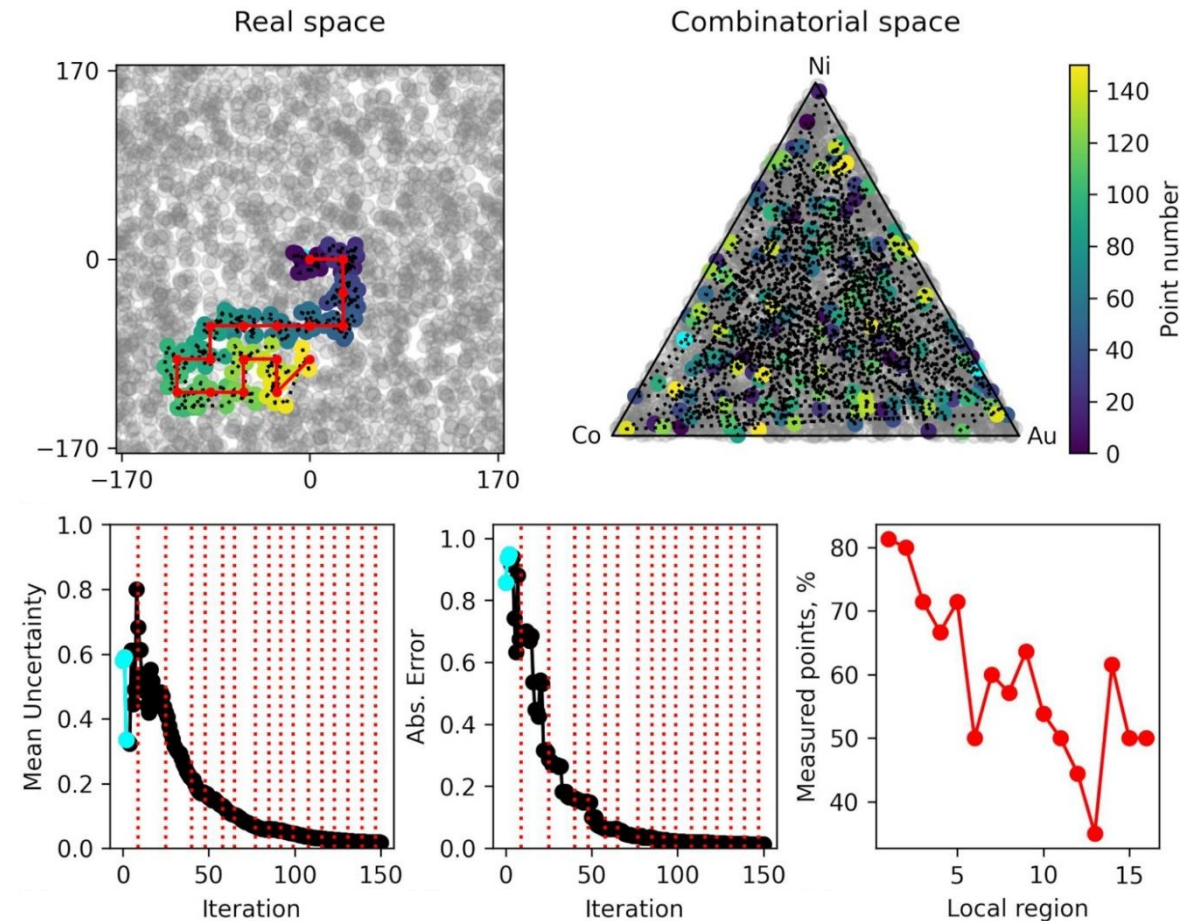
$$E_{\text{gain}}(i) = \mathbb{E}[G_{\text{new}}(i)]$$

$$V_{\text{switch}}(t) = E_{\text{gain}}(t) - C_{\text{switch}}(t)$$

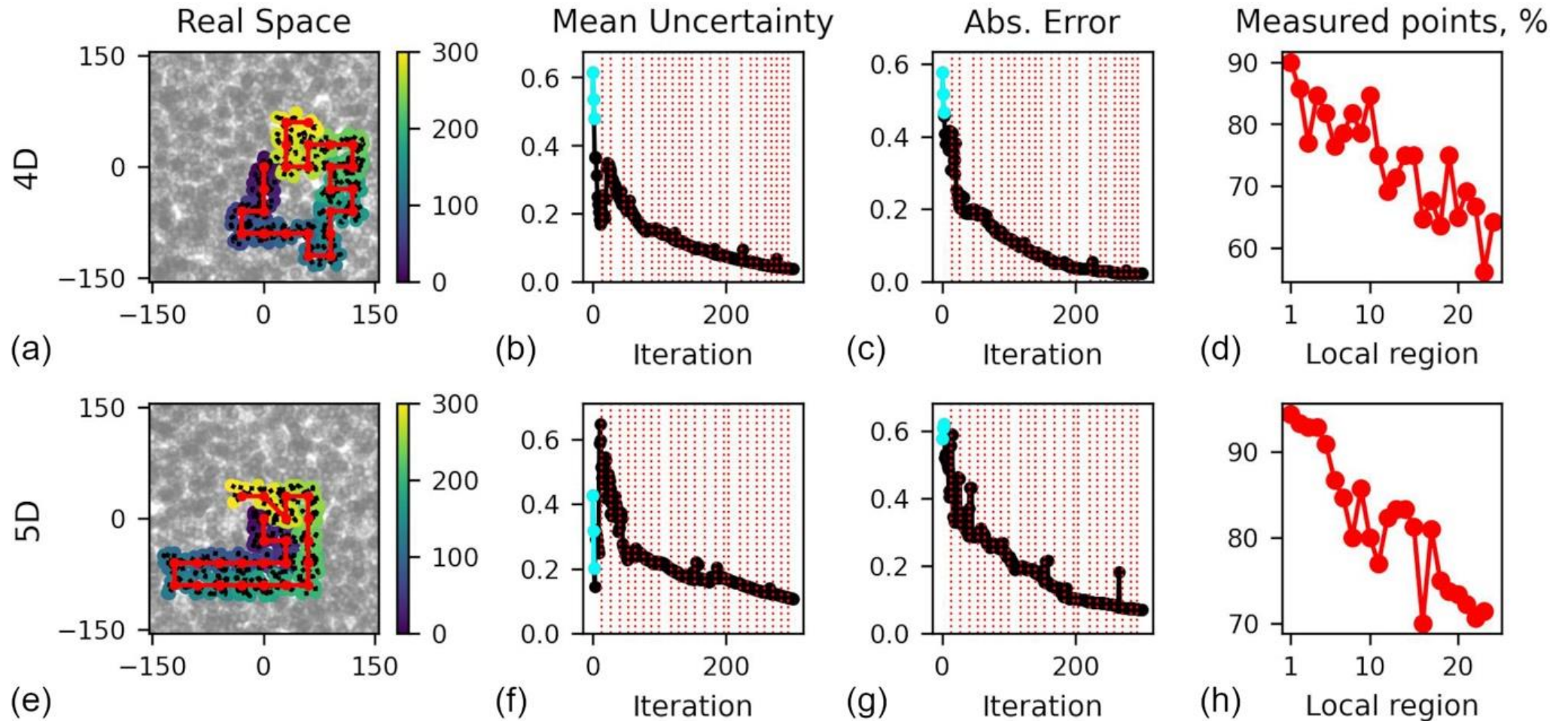
Algorithm changes region when $V_{\text{switch}} > 0$

Simulation: STEM for Random Libraries

- Autonomous exploration proceeds through spatially localized regions, with trajectories reflecting cost-aware stage motion in real space.
- Despite spatial constraints, the algorithm efficiently **samples the full ternary composition space** accessible within the random library.
- Mean predictive **uncertainty and absolute error decrease monotonically** with iteration, indicating effective information gain during exploration.
- The **fraction** of particles measured per local region **decreases** over time, **reflecting increasing model selectivity** and diminishing local opportunity.

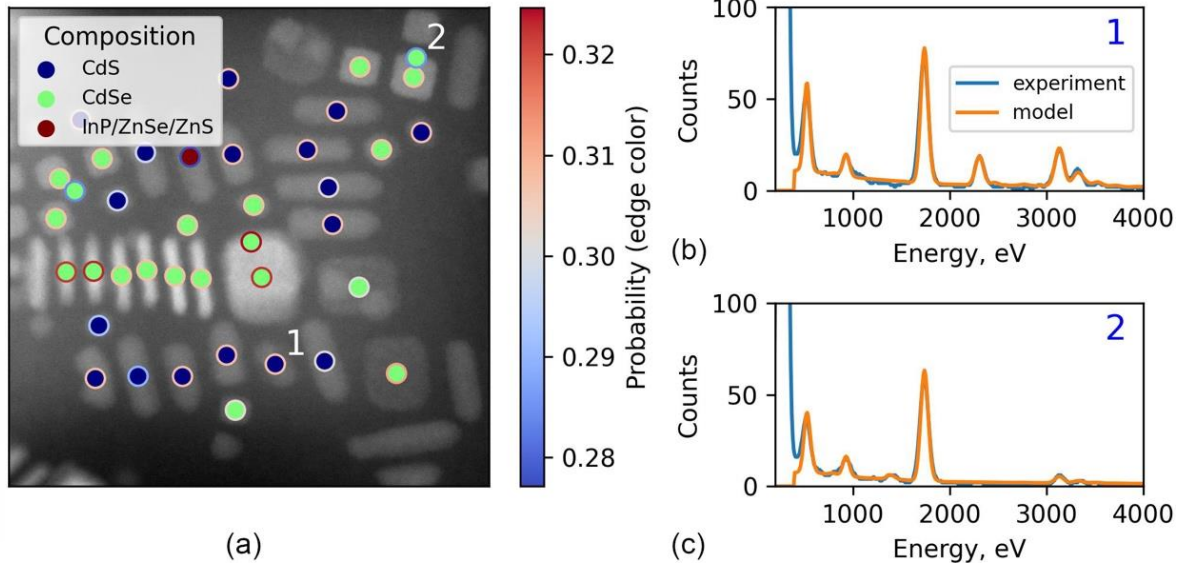


Scales to Higher Dimensions



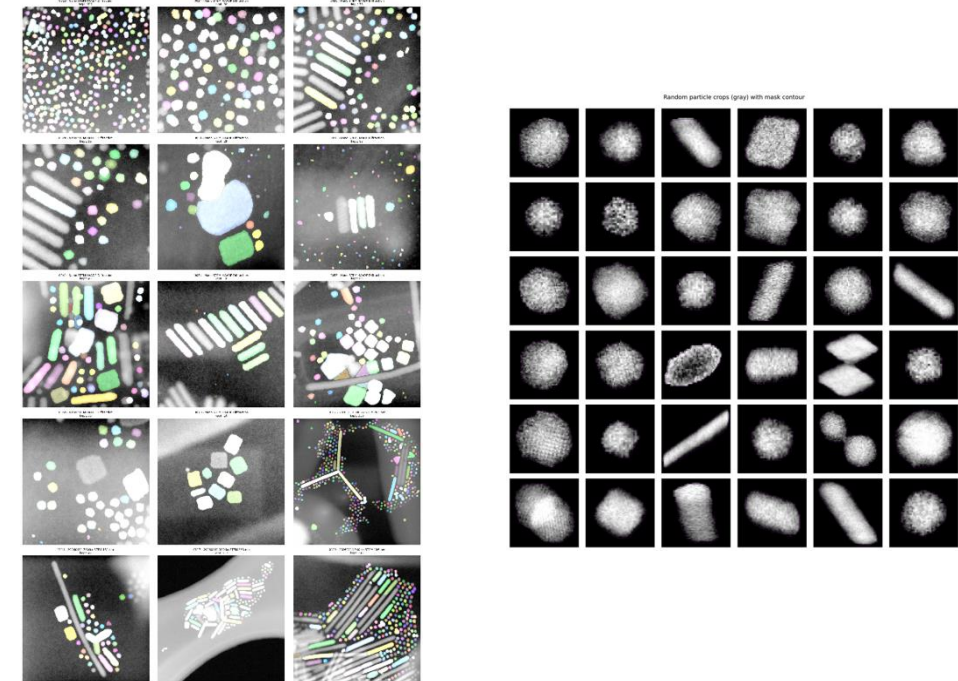
Compositional fingerprints

EDS



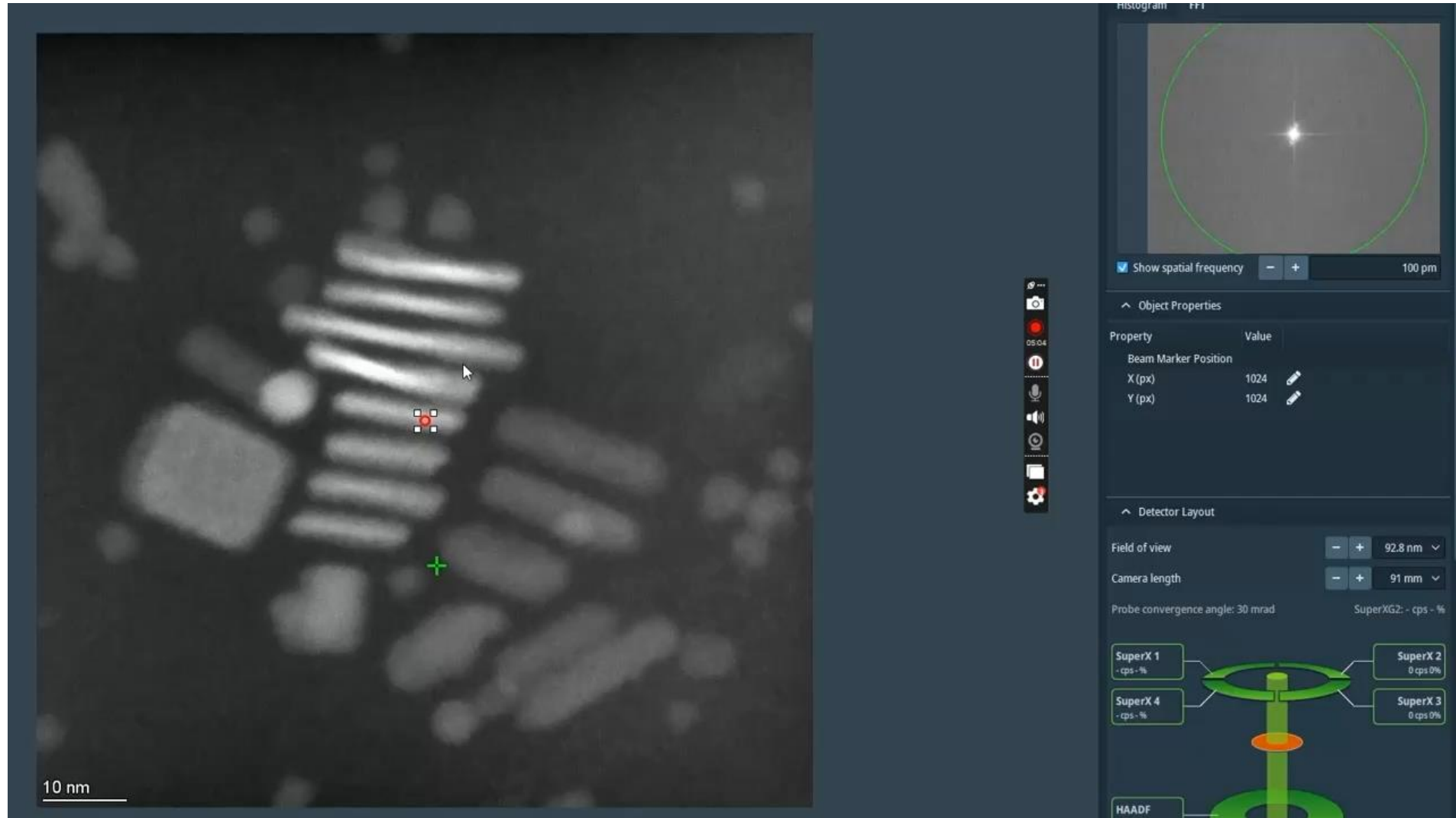
- Measuring in the selected regions

Shape



- Particle morphology can be use as complimentary descriptor

Novelty Based Exploration in STEM



Pratiush, U., Barakati, K., Slautin, B. N., Bodinger, C. C., Lowe, C. D., Cossairt, B. M., & Kalinin, S. V. (2026). Novelty-Driven Target-Space Discovery in Automated Electron and Scanning Probe Microscopy. arXiv.Org.

